

Receptors for linear acceleration:

- The Otoconia are displaced during linear acceleration to a direction opposite to the direction of movement due to their greater inertia.
- Example: when the body is suddenly moved forward → backward displacement of the Otoconia → bending backward of cilia → stimulation of groups of hair cells → the feeling of falling backwards → the person tries to correct himself by leaning forward.
- Receptors for labyrinthine righting reflexes according to head position.
- Receptors for labyrinthine statonic reflexes.
- Receptors for maintenance of posture during diving when both the visual and proprioceptive impulses are absent.

13. Physiology of basal ganglia

THE BASAL GANGLIA

The BG are nuclear masses that lie between the CC and the diencephalon at the base of the brain.

Structure: they include:

- The caudate nucleus (CN).
- The lentiform nucleus (lateral putamen and medial globus Pallidus).
- The functionally related nuclei; (RN, subthalamus and SN).

N.B. :

- The corpus striatum (CS) is formed of the CN and putamen.
- The GP is formed of external GP and internal GP parts.

- The “basal ganglia” refers to a group of subcortical nuclei within the brain responsible primarily for motor control, as well as other roles such as motor

learning, executive functions, emotional behaviours, and play an important role in reward and reinforcement, addictive behaviours and habit formation.

- The basal ganglia are located at the base of the forebrain (cerebrum) and have attracted attention in medicine for various disturbances that appear with dysfunctions caused by diseases or trauma. Disruption of the basal ganglia network forms the basis for several movement disorders eg Parkinson's Disease, Huntington Disease.

Connections of the BG: (figure 40)

(1) With the CC:

From the motor area of CC (mainly area 6) → striatum → GP (general discharge zone of ansa lenticularis) → thalamus (ventral and ventromedial nuclei) → back to motor area of the CC.

General characters of this connection:

- The output neurons of the CC, subthalamus nucleus (STN), and thalamus are excitatory secreting glutamate and Ach (in the striatum).
- The output neurons of CS and GP (GP_e and GP_i) are inhibitory secreting GABA.
- The circuit is actually formed of 2 functional circuits:

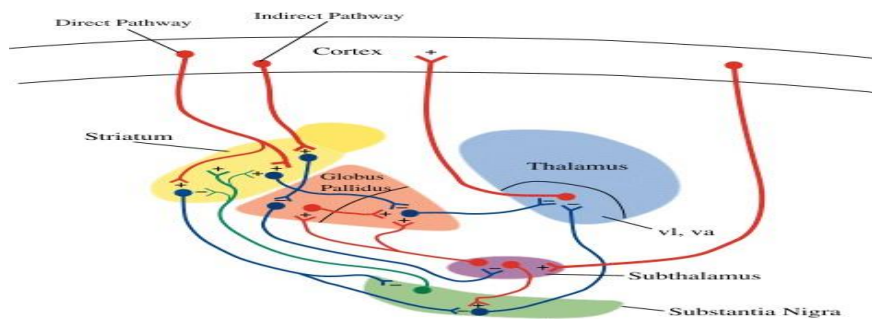


Figure 40: circuits of basal ganglia

i. The direct circuit (excitatory motor):

Motor areas of CC (stimulates - glutamate $\rightarrow$ Striatum (inhibits) - GABA $\rightarrow$ GP_i (inhibits) - GABA $\rightarrow$ thalamus ($\uparrow$ stimulation) - glutamate $\rightarrow$ CC ($\uparrow$ stimulation) $\rightarrow$ muscles (hyperkinetic state).

N.B.: inhibition by striatum of the inhibition by the (GP_i) on the thalamus increases its stimulation of the cortex resulting in hyperkinesia.

ii. The indirect circuit (inhibitory motor):

From the motor area of CC (stimulates) - glutamate $\rightarrow$ striatum (inhibits) - GABA $\rightarrow$ GP_e (inhibits) - GABA $\rightarrow$ subthalamus ($\uparrow$ stimulation) - glutamate $\rightarrow$ GP_i (inhibits) ($\uparrow$ inhibition) - GABA $\rightarrow$ thalamus ($\downarrow$ stimulation) - glutamate $\rightarrow$ CC ($\downarrow$ stimulation) -

glutamate → muscles (hypokinetic state) (figure 41). *N.B.: inhibition by striatum of the inhibition by the (GPe) on the subthalamus increases its stimulation of (GPi) and increases the inhibition of (GPi) on the thalamus, thus decreasing the stimulating effect on the cortex resulting in hypokinesia.*

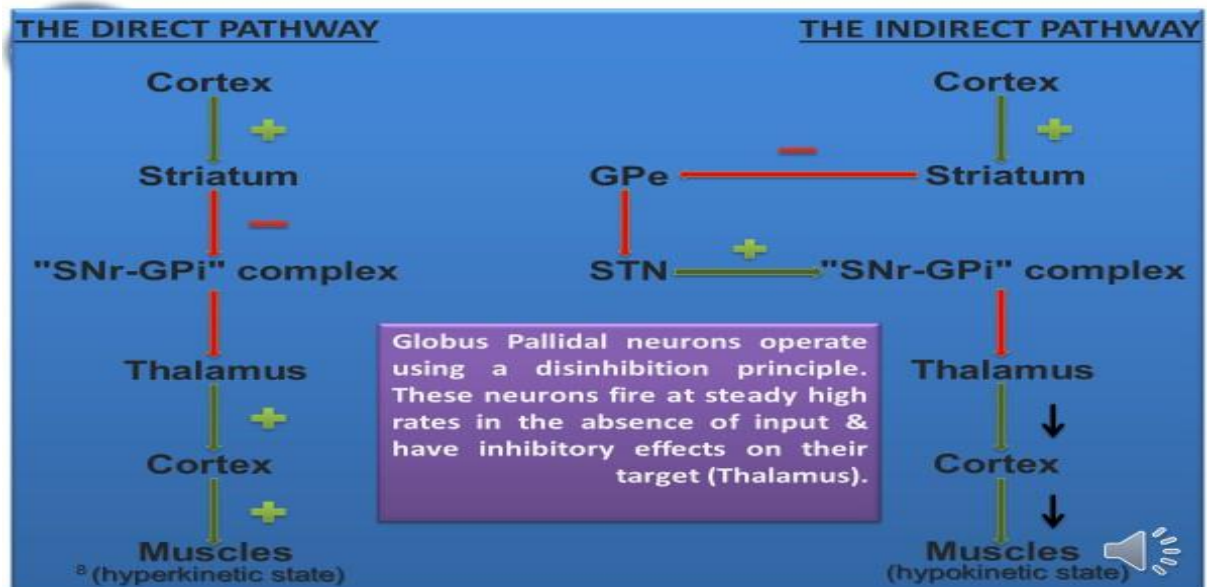


Figure 41: The direct and indirect pathways

Significance :

It is a feedback control of the motor function of the CC.

It meets the feedback circuit from the cerebellum (cortico-ponto- dentato-thalamo-cortical) at the thalamus (VL and VA nuclei) where integration occur to finely control cortical motor function.

(2) With the SN (nigrostriatal fibers):

There are mutual connections between the corpus striatum (CS) and substantia nigra (SN).

- Neurons of the SN (pars compacta, SN_c) send their axons to terminate on neurons of the CS and secrete dopamine.
- Dopamine stimulates D₁ receptors in the CS → stimulation of GABAergic neurons of the direct circuit → stimulate motor function.

- Dopamine stimulates D₂ in the CS → inhibits GABAergic neurons in the indirect circuit → ↓ inhibition of motor function.
- A balance between D1 and D2 receptor stimulation is essential for coordination of motor function.

Neurotransmitters of the BG:

- Dopamine +: From SN to Putamen and Caudate nucleus,
- GABA -: From these nuclei to Globus pallidus and SN
- Acetylcholine: From cortex to caudate nucleus and putamen ACh inhibits striatal cells that project into the

Direct Pathway

ACh cells are excitatory on striatal cells projecting into the Indirect Pathway

Other neurotransmitters such as Serotonin & Enkephalin are also present and may act as co-transmitters

Several NTs (NA, Serotonin or 5-hydroxytryptamine (5-HT), Enk) from the brain stem

Multiple excitatory glutamate pathways (not shown) that balance the inhibitory effects of GABA, Dopamine and 5HT.

FUNCTIONS OF BASAL GANGLIA

1. Control of movements
2. Planning and programming of movements
3. Cognition

Metabolic characteristics of BG

- High Oxygen consumption .

- High Copper content in Wilson's disease (Copper intoxication):
- Autosomal Recessive
- Copper binding protein Ceruloplasmin is low
- Lenticular degeneration occurs

CLINICAL APPLICATIONS (Basal ganglia disease) (table 17):

1. Hypokinetic disorders: are movement disorders that are described as having reduced motor function. This is generally attributed to higher than normal basal ganglia output causing inhibition of thalamocortical motor neurons.

Parkinsonism (figure 42)

The cardinal motor symptoms of Parkinson's disease are:

- Muscle rigidity.
- Tremor at rest (static tremors).
- Slowness in initiation and execution of movement.

Cause: Reduction in dopaminergic activity in the basal ganglia motor areas, particularly the putamen, due to gradually reduced innervation from the pars compacta of substantia nigra.

Other manifestations of Parkinsonism are:

- Autonomic dysfunction.
- Cognitive impairment
- Gait/balance difficulties.

These manifestations are thought to result from widespread progressive pathological changes commencing in the lower brain stem and ascending to the

midbrain, amygdala, thalamus and ultimately the cerebral cortex.



Figure 42: parkinsonism

2. Hyperkinetic disorders : are movement disorders characterized by increased uncontrollable motor function. They are caused by reduced basal ganglia output, which causes increased thalamocortical function which leads to the inability to stop unwanted movement.

Huntington's disease

Huntington's disease is a hereditary disease that causes defects in behavior, cognition, and uncontrolled rapid, jerky movements. Huntington's disease stems from a defect that consists of an expanded CAG repeat in a gene located on chromosome 4p. Evidence shows that the basal ganglia in patients with Huntington's Disease show a decrease in activity of the mitochondrial pathway, complex II-III. Such deficiencies are often associated with basal ganglia degeneration. This degeneration of striatal neurons projecting to GPe leads to disinhibition of the indirect pathway, increased inhibition of STN, and therefore, reduced output of the basal ganglia. The neuronal degeneration eventually causes death within 10 to 20 years.

Dystonia

Dystonia is a hyperkinetic movement disorder that is characterized by involuntary movement and the slowing of intentional movement. Though there are known causes of dystonia such as metabolic, vascular, and structural abnormalities, there are still patients with dystonia with no apparent cause. Dystonia can occur as a hyperkinetic disorder or as a side effect of hypokinetic disorders such as Parkinson's disease. Until recently it was thought that dystonia was likely caused by extreme lack of function of the direct pathway between the Putamen and the GPi. Again, it was thought that this dysfunction lead to a decrease in basal ganglia output to the thalamus and a resultant increased disinhibition of the thalamic projections to the premotor and motor cortex. However recent models in mice show that the dysfunction in the cerebellum may play an equal part in dystonia.

3. Other basal ganglia diseases: are movement disorders that generally involve the basal ganglia which do not clearly fit into being either hypo- or hyperkinetic as:
 - Tourette syndrome/obsessive–compulsive disorder
 - Sydenham's chorea

Table 17: movement disorders in basal ganglia disease

Movement Disorder	Features	Lesion
Chorea	Multiple quick, random movements, usually most prominent in the appendicular muscles	Atrophy of the striatum. Huntington Chorea
Athetosis	Slow writhing movements, which are usually more severe in the appendicular muscles	Diffuse hypermyelination of corpus striatum and thalamus

Hemiballismus	Wild flinging movements of half of the body	Hemorrhagic destruction of contralateral subthalamic n. Hypertensive patients
Parkinsonism	Pill rolling tremor of the fingers at rest, lead pipe rigidity and akinesia	Degenration of Substantia Nigra